The Jacobi identity for the Lie–Poisson bracket on the regular dual of $\mathfrak{sdiff}(\mathbb{T}^3)$, with every cancellation displayed

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Abstract

We give a complete, self-contained derivation of the Jacobi identity for the Lie–Poisson bracket $\{F, G\}(m) = \langle m, [\delta F/\delta m, \delta G/\delta m]\rangle$ on the regular dual of the Lie algebra $\mathfrak{g} = \mathfrak{sdiff}(\mathbb{T}^3)$ of smooth divergence-free vector fields on the flat torus $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$. The point of the note is the step that is usually summarised as "higher-order terms cancel": we compute the functional derivative of $\{G, H\}$ exactly, $\delta\{G, H\}/\delta m = [g, h] + \mathrm{Hess}_G(m)(K_h m) - \mathrm{Hess}_H(m)(K_g m)$, expand $\{F, \{G, H\}\} + \{G, \{H, F\}\} + \{H, \{F, G\}\}$ completely, display all six second-derivative terms, and show that they cancel in three pairs by the symmetry of second derivatives (Schwarz), leaving $\langle m, (\text{vector-field Jacobiator})\rangle = 0$. The vector-field Jacobi identity itself is proved by writing out all twelve terms. We also derive the vorticity form $\{F, G\} = \int \omega \cdot (v_F \times v_G)\,\mathrm{d}x$, explain the role of harmonic (constant) fields, and correct an integration-by-parts step from an earlier informal draft by an explicit counterexample. Symbolic (sympy) and exact-quadrature floating-point checks, and a Lean 4/mathlib formalization of *only* the final algebraic cancellation, are included. This is a classical theorem (Arnold 1966; Marsden–Weinstein 1983); nothing here is new mathematics, and nothing here bears on the Navier–Stokes global regularity problem.

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1 Purpose and what is corrected

An earlier informal draft (only an excerpt of which was available to the author of this note) asserted the Jacobi identity for the Lie–Poisson bracket on $\mathfrak{sdiff}(\mathbb{T}^3)^*$ and justified it by writing $\delta\{G,H\}/\delta\omega = v_G \times v_H + (\text{higher-order terms})$ and then letting those terms cancel "by symmetry". It also used the identity $\int \omega \cdot ((v_G \cdot \nabla)v_F)\,\mathrm{d}x = \int \omega \cdot (v_F \times v_G)\,\mathrm{d}x$, which is false (Section 8). This note:

1. fixes the setting on the genuine torus $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ (not $\mathbb{R}^3$) and treats the constant (harmonic) vector fields explicitly (Section 2);
2. proves closure and the Jacobi identity of the vector-field bracket term by term (Section 3);
3. makes the class of functionals precise and proves symmetry of their second derivatives (Section 4);
4. computes $\delta\{G,H\}/\delta m$ exactly, with *no* omitted terms (Section 6);
5. expands the cyclic sum and displays the six second-derivative terms and their pairwise cancellation (Section 7);
6. derives the correct vorticity formula and corrects the draft's step (Section 8).

2 Setting: the torus, divergence-free fields, harmonic fields

Torus and integration by parts. Let $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ with coordinates $x = (x_1, x_2, x_3)$ and Lebesgue measure $\mathrm{d}x$ (total volume $(2\pi)^3$). Smooth functions on $\mathbb{T}^3$ are smooth 2π-periodic functions on $\mathbb{R}^3$; each has a Fourier series $f(x) = \sum_{k\in\mathbb{Z}^3} \hat{f}(k)e^{ik\cdot x}$ with rapidly decreasing coefficients. Summation over repeated indices $i, j, k, l, \ldots \in \{1, 2, 3\}$ is used throughout and $\partial_j = \partial/\partial x_j$. For smooth f, h on $\mathbb{T}^3$,

$$\int_{\mathbb{T}^3} (\partial_j f)\, h \,\mathrm{d}x = -\int_{\mathbb{T}^3} f\, \partial_j h \,\mathrm{d}x, \tag{1}$$

because $\int_0^{2\pi} \partial_j(fh)\, dx_j = 0$ by periodicity (no boundary terms). In particular $\int_{\mathbb{T}^3} \operatorname{div} c \,\mathrm{d}x = 0$ for every smooth vector field c.

The Lie algebra.

$$\mathfrak{g} := \{v \in C^\infty(\mathbb{T}^3;\mathbb{R}^3) :\ \operatorname{div} v = \partial_j v_j = 0\}.$$

We do not impose zero mean on $\mathfrak{g}$: constant vector fields belong to $\mathfrak{g}$. $\mathfrak{g}$ carries the Fréchet topology of C^∞ (uniform convergence of all derivatives), with norms $\|v\|_{C^r} = \max_{|\alpha|\le r,\, i} \sup |\partial^\alpha v_i|$. Let $\mathfrak{g}_0 := \{v \in \mathfrak{g} : \int_{\mathbb{T}^3} v \,\mathrm{d}x = 0\}$.

Lemma 2.1 (Helmholtz–Leray decomposition on $\mathbb{T}^3$)**.** *Every $m \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ decomposes uniquely as $m = Pm + \nabla p$ with $Pm \in \mathfrak{g}$ and $p \in C^\infty(\mathbb{T}^3)$ (unique up to an additive constant). In Fourier variables $\widehat{Pm}(0) = \hat m(0)$ and, for $k \neq 0$, $\widehat{Pm}(k) = \hat m(k) - k\,\frac{k\cdot\hat m(k)}{|k|^2}$, $\hat p(k) = -i\frac{k\cdot\hat m(k)}{|k|^2}$. The decomposition is L^2-orthogonal: $\int v\cdot\nabla p\,\mathrm{d}x = 0$ for $v \in \mathfrak{g}$. Consequently $\int v\cdot m\,\mathrm{d}x = \int v\cdot Pm\,\mathrm{d}x$ for all $v \in \mathfrak{g}$, and P is symmetric: $\int a\cdot Pb\,\mathrm{d}x = \int Pa\cdot b\,\mathrm{d}x$.*

Proof. The displayed formulas give $k\cdot\widehat{Pm}(k) = 0$, i.e. $\operatorname{div} Pm = 0$, and $ik\hat p(k) = k(k\cdot\hat m(k))/|k|^2$, so $\hat m = \widehat{Pm} + \widehat{\nabla p}$. The coefficients are bounded by $|\hat m(k)|$, so rapid decay (smoothness) is preserved. Orthogonality: by (1), $\int v_j\partial_j p\,\mathrm{d}x = -\int(\partial_j v_j)p\,\mathrm{d}x = 0$. Uniqueness: if $v + \nabla p = 0$ with $v \in \mathfrak{g}$ then $0 = \int(v+\nabla p)\cdot\nabla p = \int|\nabla p|^2$, so $\nabla p = 0$, $v = 0$. Symmetry: $\int a\cdot Pb = \int Pa\cdot Pb = \int Pa\cdot b$ by orthogonality. □

Lemma 2.2 (Harmonic fields; curl and Biot–Savart on $\mathbb{T}^3$)**.** *For $v \in \mathfrak{g}$: $\operatorname{curl} v = 0$ if and only if v is constant. The map $\operatorname{curl} : \mathfrak{g} \to \mathfrak{g}_0$ is onto with kernel the constant fields $\mathbb{R}^3$, and on $\mathfrak{g}_0$ it has the inverse (Biot–Savart law) $\operatorname{curl}^{-1}\omega = \operatorname{curl}(-\Delta)^{-1}\omega$, where $(-\Delta)^{-1}$ acts on mean-zero fields by $\hat\omega(k) \mapsto \hat\omega(k)/|k|^2$. Every $m \in \mathfrak{g}$ is $m = \bar m + \operatorname{curl}(-\Delta)^{-1}\operatorname{curl} m$ with $\bar m = (2\pi)^{-3}\int m\,\mathrm{d}x$.*

Proof. $\widehat{\operatorname{curl} v}(k) = ik\times\hat v(k)$. If $\operatorname{div} v = 0$ then $k \perp \hat v(k)$, so $|k\times\hat v(k)| = |k|\,|\hat v(k)|$; hence $\operatorname{curl} v = 0$ forces $\hat v(k) = 0$ for $k \neq 0$, i.e. v is constant (and constants have zero curl). $\operatorname{curl} v$ is divergence free and has $\widehat{\operatorname{curl} v}(0) = 0$, so $\operatorname{curl}\mathfrak{g} \subset \mathfrak{g}_0$. For $v \in \mathfrak{g}$, the identity $\operatorname{curl}\operatorname{curl} v = \nabla\operatorname{div} v - \Delta v = -\Delta v$ (check on Fourier modes: $ik\times(ik\times\hat v) = -k(k\cdot\hat v) + |k|^2\hat v$) shows that for $\omega \in \mathfrak{g}_0$, $v := \operatorname{curl}(-\Delta)^{-1}\omega \in \mathfrak{g}_0$ satisfies $\operatorname{curl} v = -\Delta(-\Delta)^{-1}\omega = \omega$ (note $\operatorname{div}(-\Delta)^{-1}\omega = 0$). The last statement follows since $m - \bar m \in \mathfrak{g}_0$. □

Remark 2.3. The constant fields are exactly the harmonic vector fields of the flat torus ($H^1_{\mathrm{dR}}(\mathbb{T}^3) \cong \mathbb{R}^3$). Vorticity $\omega = \operatorname{curl} m$ therefore does not determine m: the mean $\bar m$ is lost. Any formula with "curl^{-1}" on $\mathbb{T}^3$ must either restrict to mean-zero fields or carry $\bar m$ separately. On $\mathbb{R}^3$ this issue is replaced by decay conditions; the two settings are different.

3 The Lie algebra $\mathfrak{sdiff}(\mathbb{T}^3)$

Definition 3.1 (Bracket and sign convention)**.** For smooth vector fields v, w set

$$[v,w] := (w\cdot\nabla)v - (v\cdot\nabla)w, \qquad [v,w]_i = w_j\partial_j v_i - v_j\partial_j w_i. \tag{2}$$

With $\mathcal{L}_X f := X_j\partial_j f$ (derivative of a function f along X) this reads $[v,w]_i = \mathcal{L}_w v_i - \mathcal{L}_v w_i$. It is the *negative* of the commutator bracket $[v,w]_{\mathrm{VF}} := (v\cdot\nabla)w - (w\cdot\nabla)v$ of vector fields as derivations; this is the convention in which $[\cdot,\cdot]$ is the Lie algebra bracket of the group of (volume-preserving) diffeomorphisms (Arnold). The negative of a Lie bracket is again a Lie bracket, so both conventions give Lie algebras; the sign of the Poisson bracket is fixed in Proposition 5.3.

Lemma 3.2. *For all smooth vector fields a, b: $\operatorname{curl}(a\times b) = a\,\operatorname{div} b - b\,\operatorname{div} a + (b\cdot\nabla)a - (a\cdot\nabla)b$.*

Proof. With $(a\times b)_k = \varepsilon_{klm}a_l b_m$, $(\operatorname{curl} c)_i = \varepsilon_{ijk}\partial_j c_k$, and $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$:

$$\begin{aligned}\operatorname{curl}(a\times b)_i &= \varepsilon_{ijk}\varepsilon_{klm}\partial_j(a_l b_m) = \partial_m(a_i b_m) - \partial_l(a_l b_i)\\ &= a_i\,\partial_m b_m + b_m\partial_m a_i - b_i\,\partial_l a_l - a_l\partial_l b_i = \big(a\operatorname{div} b + (b\cdot\nabla)a - b\operatorname{div} a - (a\cdot\nabla)b\big)_i.\end{aligned}$$ □

Corollary 3.3. *If $\operatorname{div} v = \operatorname{div} w = 0$ then $[v,w] = (w\cdot\nabla)v - (v\cdot\nabla)w = \operatorname{curl}(v\times w)$. (Without $\operatorname{div} v = \operatorname{div} w = 0$ this fails: for $v = (\sin x_1, 0, 0)$, $w = (0, \cos x_1, 0)$, $[v,w] - \operatorname{curl}(v\times w) = (0, \cos^2 x_1, 0)$.)*

Proof. Lemma 3.2 with $a = v$, $b = w$; the divergence terms vanish and the remaining terms are exactly (2). □

Proposition 3.4 (Closure)**.** *If $v, w \in \mathfrak{g}$ then $[v, w] \in \mathfrak{g}$; moreover $\int [v, w]\,\mathrm{d}x = 0$, so $[\mathfrak{g}, \mathfrak{g}] \subset \mathfrak{g}_0$.*

Proof. Directly:

$$\partial_i [v, w]_i = \partial_i (w_j \partial_j v_i - v_j \partial_j w_i) = (\partial_i w_j)(\partial_j v_i) + w_j \partial_j (\partial_i v_i) - (\partial_i v_j)(\partial_j w_i) - v_j \partial_j (\partial_i w_i).$$

The second and fourth terms vanish since $\partial_i v_i = \partial_i w_i = 0$. Renaming the summation indices $i \leftrightarrow j$ in the third term turns $(\partial_i v_j)(\partial_j w_i)$ into $(\partial_j v_i)(\partial_i w_j)$, which equals the first term; so the first and third cancel. Smoothness is clear. Zero mean: $[v, w] = \operatorname{curl}(v \times w)$ by Corollary 3.3, and $\int \operatorname{curl} c\,\mathrm{d}x = 0$ by (1). □

Lemma 3.5 (Commutator of derivations)**.** *For smooth vector fields X, Y (not necessarily divergence free) and smooth f, $\mathcal{L}_X \mathcal{L}_Y f - \mathcal{L}_Y \mathcal{L}_X f = \mathcal{L}_{[X,Y]_{\mathrm{VF}}} f$, hence $\mathcal{L}_{[X,Y]} f = \mathcal{L}_Y \mathcal{L}_X f - \mathcal{L}_X \mathcal{L}_Y f$.*

Proof.

$$\begin{aligned} \mathcal{L}_X \mathcal{L}_Y f - \mathcal{L}_Y \mathcal{L}_X f &= X_j \partial_j (Y_k \partial_k f) - Y_j \partial_j (X_k \partial_k f) \\ &= X_j (\partial_j Y_k) \partial_k f + X_j Y_k\, \partial_j \partial_k f - Y_j (\partial_j X_k) \partial_k f - Y_j X_k\, \partial_j \partial_k f. \end{aligned}$$

Rename $j \leftrightarrow k$ in the fourth term: $Y_j X_k \partial_j \partial_k f = Y_k X_j \partial_k \partial_j f = X_j Y_k \partial_j \partial_k f$ (equality of mixed partials of the smooth f). So the second and fourth terms cancel and what remains is $(\mathcal{L}_X Y_k - \mathcal{L}_Y X_k) \partial_k f = \mathcal{L}_{[X,Y]_{\mathrm{VF}}} f$. Finally $[X, Y] = -[X, Y]_{\mathrm{VF}}$. □

Proposition 3.6 (Jacobi identity for the vector-field bracket)**.** *For all smooth vector fields u, v, w (divergence-free or not), $[u, [v, w]] + [v, [w, u]] + [w, [u, v]] = 0$.*

Proof. Fix a component i. By (2) with $X = [v, w]$, $[u, [v, w]]_i = \mathcal{L}_{[v,w]} u_i - \mathcal{L}_u([v, w]_i)$. By Lemma 3.5 (applied to $f = u_i$), $\mathcal{L}_{[v,w]} u_i = \mathcal{L}_w \mathcal{L}_v u_i - \mathcal{L}_v \mathcal{L}_w u_i$, and $\mathcal{L}_u([v, w]_i) = \mathcal{L}_u \mathcal{L}_w v_i - \mathcal{L}_u \mathcal{L}_v w_i$. Thus

$$[u, [v, w]]_i = \underbrace{\mathcal{L}_w \mathcal{L}_v u_i}_{(1)}\ \underbrace{-\mathcal{L}_v \mathcal{L}_w u_i}_{(2)}\ \underbrace{-\mathcal{L}_u \mathcal{L}_w v_i}_{(3)}\ \underbrace{+\mathcal{L}_u \mathcal{L}_v w_i}_{(4)},$$

and by the cyclic substitution $u \to v \to w \to u$,

$$\begin{aligned} [v, [w, u]]_i &= \underbrace{\mathcal{L}_u \mathcal{L}_w v_i}_{(3')}\ \underbrace{-\mathcal{L}_w \mathcal{L}_u v_i}_{(5)}\ \underbrace{-\mathcal{L}_v \mathcal{L}_u w_i}_{(6)}\ \underbrace{+\mathcal{L}_v \mathcal{L}_w u_i}_{(2')}, \\ [w, [u, v]]_i &= \underbrace{\mathcal{L}_v \mathcal{L}_u w_i}_{(6')}\ \underbrace{-\mathcal{L}_u \mathcal{L}_v w_i}_{(4')}\ \underbrace{-\mathcal{L}_w \mathcal{L}_v u_i}_{(1')}\ \underbrace{+\mathcal{L}_w \mathcal{L}_u v_i}_{(5')}. \end{aligned}$$

The twelve terms cancel in the six pairs (1)+(1′), (2)+(2′), (3)+(3′), (4)+(4′), (5)+(5′), (6)+(6′):

pair	term	partner
(1)	$+\mathcal{L}_w \mathcal{L}_v u_i$ in $[u, [v, w]]$	$-\mathcal{L}_w \mathcal{L}_v u_i$ in $[w, [u, v]]$
(2)	$-\mathcal{L}_v \mathcal{L}_w u_i$ in $[u, [v, w]]$	$+\mathcal{L}_v \mathcal{L}_w u_i$ in $[v, [w, u]]$
(3)	$-\mathcal{L}_u \mathcal{L}_w v_i$ in $[u, [v, w]]$	$+\mathcal{L}_u \mathcal{L}_w v_i$ in $[v, [w, u]]$
(4)	$+\mathcal{L}_u \mathcal{L}_v w_i$ in $[u, [v, w]]$	$-\mathcal{L}_u \mathcal{L}_v w_i$ in $[w, [u, v]]$
(5)	$-\mathcal{L}_w \mathcal{L}_u v_i$ in $[v, [w, u]]$	$+\mathcal{L}_w \mathcal{L}_u v_i$ in $[w, [u, v]]$
(6)	$-\mathcal{L}_v \mathcal{L}_u w_i$ in $[v, [w, u]]$	$+\mathcal{L}_v \mathcal{L}_u w_i$ in $[w, [u, v]]$

All second derivatives were already absorbed in Lemma 3.5, where their cancellation was shown explicitly. □

Remark 3.7. The divergence-free condition is *not* used in Proposition 3.6; it is used only for closure (Proposition 3.4) and for $[v, w] = \mathrm{curl}(v \times w)$. The draft's claim that the cancellation needs $\partial_j u_j = 0$ is therefore inaccurate (though harmless).

4 The regular dual and the class of functionals

Pairing and regular dual. For smooth vector fields put $\langle m, v\rangle := \int_{\mathbb{T}^3} m \cdot v \, \mathrm{d}x$. Note $|\langle m, v\rangle| \le 3(2\pi)^3 \|m\|_{C^0} \|v\|_{C^0}$.

Proposition 4.1 (Regular dual)**.** *(i) The pairing restricted to $\mathfrak{g} \times \mathfrak{g}$ is nondegenerate: if $m \in \mathfrak{g}$ and $\langle m, v\rangle = 0$ for all $v \in \mathfrak{g}$ then $m = 0$. (ii) The "regular dual" $\mathfrak{g}^*_{\mathrm{reg}} := C^\infty(\mathbb{T}^3; \mathbb{R}^3)/\nabla C^\infty(\mathbb{T}^3)$ (momentum, or 1-forms, modulo gradients/exact forms), paired with $\mathfrak{g}$ by $\langle [m], v\rangle = \langle m, v\rangle$, is identified with $\mathfrak{g}$ by $[m] \mapsto Pm$, and $\langle [m], v\rangle = \langle Pm, v\rangle$ for $v \in \mathfrak{g}$.*

Proof. (i) Take $v = m$: $\int |m|^2 = 0$. (ii) The pairing is well defined on classes since $\langle \nabla p, v\rangle = 0$ for $v \in \mathfrak{g}$ (Lemma 2.1); $[m] \mapsto Pm$ is well defined ($P\nabla p = 0$), injective ($Pm = 0 \Rightarrow m = \nabla p$) and onto ($Pv = v$ for $v \in \mathfrak{g}$); the last identity is Lemma 2.1. □

Henceforth the phase space is $\mathfrak{g}$ itself, a point $m \in \mathfrak{g}$ being the divergence-free representative of a momentum class. (The full topological dual of $\mathfrak{g}$ contains distributions; we use only the regular part, as is standard.)

Definition 4.2 (Admissible functionals $\mathcal{A}$)**.** A function $F : \mathfrak{g} \to \mathbb{R}$ is *admissible*, $F \in \mathcal{A}$, if for all $m, a, b \in \mathfrak{g}$:

(A1) $DF(m)a := \frac{d}{d\varepsilon}\big|_{\varepsilon=0} F(m + \varepsilon a)$ exists and there is $\frac{\delta F}{\delta m}(m) \in \mathfrak{g}$ with $DF(m)a = \langle a, \frac{\delta F}{\delta m}(m)\rangle$ (unique by Proposition 4.1(i));

(A2) the curve $t \mapsto \frac{\delta F}{\delta m}(m + tb) \in \mathfrak{g}$ is differentiable at $t = 0$ in the C^∞ topology (difference quotients converge in every C^r norm); its derivative is denoted $\mathrm{Hess}_F(m)b \in \mathfrak{g}$;

(A3) $\varphi(s, t) := F(m + sa + tb)$ is of class C^2 on $\mathbb{R}^2$.

We write $Q_F(m)(a, b) := \langle a, \mathrm{Hess}_F(m)b\rangle$; this is $D^2F(m)(b, a)$.

This is a Gateaux-type, C^∞-Fréchet-space setting; no Banach-space H^s setting is used, because the bracket loses one derivative and $\mathfrak{g}$ would not be closed under it in a fixed H^s. Functionals such as a point evaluation $m \mapsto m(x_0) \cdot e$ are excluded: their derivative is a distribution, not an element of $\mathfrak{g}$.

Lemma 4.3 (Symmetry of the second derivative)**.** *If $F \in \mathcal{A}$ then $Q_F(m)(a, b) = Q_F(m)(b, a)$ for all $m, a, b \in \mathfrak{g}$.*

Proof. Let $\varphi(s, t) = F(m + sa + tb)$. By (A1) at the base point $m + sa + tb$, $\partial_s\varphi(s, t) = DF(m + sa + tb)a = \langle a, \frac{\delta F}{\delta m}(m + sa + tb)\rangle$. Setting $s = 0$, $\partial_s\varphi(0, t) = \langle a, \frac{\delta F}{\delta m}(m + tb)\rangle$, so

$$\partial_t\partial_s\varphi(0, 0) = \lim_{t\to 0} \Big\langle a, \tfrac{1}{t}\big(\frac{\delta F}{\delta m}(m + tb) - \frac{\delta F}{\delta m}(m)\big)\Big\rangle = \langle a, \mathrm{Hess}_F(m)b\rangle,$$

because by (A2) the difference quotients converge in C^0 and the pairing is C^0-continuous. Exchanging the roles of (s, a) and (t, b), $\partial_s\partial_t\varphi(0, 0) = \langle b, \mathrm{Hess}_F(m)a\rangle$. By (A3) and Schwarz's theorem for C^2 functions on $\mathbb{R}^2$, $\partial_t\partial_s\varphi(0, 0) = \partial_s\partial_t\varphi(0, 0)$. □

Remark 4.4. $a \mapsto Q_F(m)(a,b)$ is linear, hence by symmetry so is $b \mapsto Q_F(m)(a,b)$, and by nondegeneracy $b \mapsto \mathrm{Hess}_F(m)b$ is linear.

Example 4.5 (Admissible functionals)**.** (i) $F(m) = \langle m, a_0\rangle$, $a_0 \in \mathfrak{g}$: $\frac{\delta F}{\delta m} = a_0$, $\mathrm{Hess}_F = 0$. (ii) $E(m) = \frac{1}{2}\langle m, m\rangle$: $\frac{\delta E}{\delta m} = m$, $\mathrm{Hess}_E\, b = b$. (iii) Cylindrical: $F(m) = \Phi(\langle m, a_1\rangle, \dots, \langle m, a_n\rangle)$, $a_r \in \mathfrak{g}$, $\Phi \in C^2(\mathbb{R}^n)$: $\frac{\delta F}{\delta m} = \sum_r \partial_r\Phi\, a_r$, $\mathrm{Hess}_F\, b = \sum_{r,s} \partial_r\partial_s\Phi\, \langle b, a_s\rangle\, a_r$. (iv) Local: $F(m) = \int_{\mathbb{T}^3} \psi(m(x))\,\mathrm{d}x$, $\psi \in C^\infty(\mathbb{R}^3)$: $DF(m)a = \int \nabla\psi(m)\cdot a = \langle a, P(\nabla\psi \circ m)\rangle$ (Lemma 2.1, $a \in \mathfrak{g}$), so $\frac{\delta F}{\delta m} = P(\nabla\psi \circ m)$ and $\mathrm{Hess}_F\, b = P((\nabla^2\psi \circ m)\, b)$; (A2) holds because $(t,x) \mapsto \nabla\psi(m(x) + tb(x))$ is smooth on $\mathbb{R}\times\mathbb{T}^3$, so (Taylor with integral remainder applied to each ∂_x^α) its t-difference quotients converge in every C^r_x norm, and P is continuous on C^∞; (A3) is differentiation under the integral sign. Symmetry of Q_F is visible: $\langle a, P(\nabla^2\psi(m)b)\rangle = \int a\cdot\nabla^2\psi(m)b\,\mathrm{d}x$.

5 Coadjoint representative and the bracket

Lemma 5.1 (Coadjoint representative)**.** *For $m, v, w \in \mathfrak{g}$ with $\omega := \operatorname{curl} m$,*

$$\langle m, [v,w]\rangle = \int_{\mathbb{T}^3} \omega\cdot(v\times w)\,\mathrm{d}x = \langle K_w m, v\rangle, \qquad K_w m := P(w\times\omega) \in \mathfrak{g}.$$

Proof. First, for smooth A, B: $\operatorname{div}(A\times B) = \partial_i(\varepsilon_{ijk}A_jB_k) = B_k\varepsilon_{kij}\partial_iA_j - A_j\varepsilon_{jik}\partial_iB_k = B\cdot\operatorname{curl}A - A\cdot\operatorname{curl}B$. With $A = c := v\times w$ and $B = m$, integrating and using $\int\operatorname{div} = 0$: $\int m\cdot\operatorname{curl} c\,\mathrm{d}x = \int(\operatorname{curl} m)\cdot c\,\mathrm{d}x$. Since $[v,w] = \operatorname{curl}(v\times w)$ (Corollary 3.3), $\langle m,[v,w]\rangle = \int\omega\cdot(v\times w)\,\mathrm{d}x$. The scalar triple product is cyclic: $\omega\cdot(v\times w) = v\cdot(w\times\omega)$. Finally $\int v\cdot(w\times\omega) = \langle v, P(w\times\omega)\rangle$ since $v\in\mathfrak{g}$ (Lemma 2.1). □

Thus $v \mapsto \langle m,[v,w]\rangle$ is represented in $\mathfrak{g}$ by $K_w m$ (this is $\pm\mathrm{ad}^*_w m$ depending on conventions; we only use the defining property $\langle K_w m, v\rangle = \langle m,[v,w]\rangle$). Note $K_w m = 0$ when m is constant.

Definition 5.2 (Lie–Poisson bracket)**.** For $F, G \in \mathcal{A}$ and $m \in \mathfrak{g}$,

$$\{F,G\}_\pm(m) := \pm\Big\langle m, \Big[\frac{\delta F}{\delta m}(m), \frac{\delta G}{\delta m}(m)\Big]\Big\rangle.$$

It is bilinear and antisymmetric.

Proposition 5.3 (Sign: Euler's equations)**.** *Let $F_a(m) = \langle m, a\rangle$ ($a \in \mathfrak{g}$) and $E(m) = \frac{1}{2}\langle m,m\rangle$. Then $\{F_a, E\}_+(m) = -\langle (m\cdot\nabla)m, a\rangle$. Hence "$\frac{d}{dt}F(m) = \{F,E\}_+$ for all F" is equivalent to $\partial_t m = -P\big((m\cdot\nabla)m\big)$, the incompressible Euler equations $\partial_t u + (u\cdot\nabla)u = -\nabla p$, $\operatorname{div} u = 0$, with $u = m$.*

Proof. $\{F_a,E\}_+ = \langle m,[a,m]\rangle = \int m_i(m_j\partial_j a_i - a_j\partial_j m_i)\,\mathrm{d}x$. Second part: $\int m_i a_j\partial_j m_i = \frac{1}{2}\int a_j\partial_j|m|^2 = -\frac{1}{2}\int(\partial_j a_j)|m|^2 = 0$. First part:

$$\int m_i m_j\partial_j a_i = -\int\partial_j(m_im_j)a_i = -\int(m_j\partial_j m_i + m_i\partial_j m_j)a_i = -\int((m\cdot\nabla)m)\cdot a.$$

Testing against all $a\in\mathfrak{g}$ and using Lemma 2.1 gives the Leray form. □

So with the bracket convention (2), the "+" bracket generates Euler's equations; "−" corresponds to the opposite (left/right) convention. The Jacobi identity holds for both (Theorem 7.1).

6 The exact functional derivative of $\{G, H\}$

Lemma 6.1 (Product rule)**.** *Let $m, a \in \mathfrak{g}$ and let $g(\varepsilon), h(\varepsilon)$ be curves in $\mathfrak{g}$ differentiable at $\varepsilon = 0$ in the C^∞ topology, with derivatives g', h'. Then $T(\varepsilon) := \langle m + \varepsilon a, [g(\varepsilon), h(\varepsilon)]\rangle$ satisfies $T'(0) = \langle a, [g, h]\rangle + \langle m, [g', h]\rangle + \langle m, [g, h']\rangle$, where $g = g(0)$, $h = h(0)$.*

Proof. By trilinearity, $T(\varepsilon) - T(0) = \varepsilon\langle a, [g(\varepsilon), h(\varepsilon)]\rangle + \langle m, [g(\varepsilon) - g, h(\varepsilon)]\rangle + \langle m, [g, h(\varepsilon) - h]\rangle$. From (2), $\|[p, q]\|_{C^0} \le 3(\|q\|_{C^0}\|p\|_{C^1} + \|p\|_{C^0}\|q\|_{C^1})$, so $(m, p, q) \mapsto \langle m, [p, q]\rangle$ is continuous for the $C^0 \times C^1 \times C^1$ norms. Divide by ε and let $\varepsilon \to 0$: $(g(\varepsilon) - g)/\varepsilon \to g'$ and $(h(\varepsilon) - h)/\varepsilon \to h'$ in C^1, and $g(\varepsilon) \to g$, $h(\varepsilon) \to h$ in C^1. □

Proposition 6.2 (Functional derivative of a bracket — no omitted terms)**.** *Let $G, H \in \mathcal{A}$, $m \in \mathfrak{g}$, $g := \frac{\delta G}{\delta m}(m)$, $h := \frac{\delta H}{\delta m}(m)$. Then $\{G, H\}_+$ satisfies (A1) and*

$$\boxed{\frac{\delta\{G, H\}_+}{\delta m}(m) = [g, h] + \mathrm{Hess}_G(m)\big(K_h m\big) - \mathrm{Hess}_H(m)\big(K_g m\big)} \tag{3}$$

with $K_w m = P(w \times \operatorname{curl} m)$ from Lemma 5.1; and $\delta\{G, H\}_-/\delta m = -\delta\{G, H\}_+/\delta m$.

Proof. Fix $a \in \mathfrak{g}$. Apply Lemma 6.1 with $g(\varepsilon) = \frac{\delta G}{\delta m}(m + \varepsilon a)$, $h(\varepsilon) = \frac{\delta H}{\delta m}(m + \varepsilon a)$, which by (A2) have derivatives $\mathrm{Hess}_G(m)a$, $\mathrm{Hess}_H(m)a$:

$$D\{G, H\}_+(m)a = \langle a, [g, h]\rangle + \langle m, [\mathrm{Hess}_G(m)a,\ h]\rangle + \langle m, [g,\ \mathrm{Hess}_H(m)a]\rangle. \tag{4}$$

Second term: by Lemma 5.1 with $v = \mathrm{Hess}_G(m)a \in \mathfrak{g}$, $w = h$, and then Lemma 4.3,

$$\begin{aligned}\langle m, [\mathrm{Hess}_G(m)a, h]\rangle &= \langle K_h m, \mathrm{Hess}_G(m)a\rangle = Q_G(m)(K_h m, a)\\ &= Q_G(m)(a, K_h m) = \langle a, \mathrm{Hess}_G(m)K_h m\rangle.\end{aligned}$$

Third term: by antisymmetry, Lemma 5.1 with $v = \mathrm{Hess}_H(m)a$, $w = g$, and Lemma 4.3,

$$\langle m, [g, \mathrm{Hess}_H(m)a]\rangle = -\langle m, [\mathrm{Hess}_H(m)a, g]\rangle = -\langle K_g m, \mathrm{Hess}_H(m)a\rangle = -\langle a, \mathrm{Hess}_H(m)K_g m\rangle.$$

Hence $D\{G, H\}_+(m)a = \langle a, [g, h] + \mathrm{Hess}_G(m)K_h m - \mathrm{Hess}_H(m)K_g m\rangle$ for all $a \in \mathfrak{g}$; all three vectors lie in $\mathfrak{g}$ (Proposition 3.4, (A2)), which is (A1) with (3). □

Remark 6.3 (What the draft's "higher-order terms" are)**.** The terms $\mathrm{Hess}_G(m)K_h m - \mathrm{Hess}_H(m)K_g m$ are the "higher-order functional derivative terms". They do *not* vanish in general: for $G = E = \frac{1}{2}\langle m, m\rangle$ and $H = F_a$ linear, (3) gives $\delta\{E, F_a\}/\delta m = [m, a] + K_a m = [m, a] + P(a \times \omega)$, and the second term is nonzero for generic m, a. The numerical test (a) in Section 9 confirms (3) against a finite-difference derivative and shows that $[g, h]$ alone is wrong by 29–93%. They disappear only in the cyclic sum, as shown next.

7 The Jacobi identity: full expansion and the six cancelling terms

Theorem 7.1 (Jacobi identity)**.** *For $F, G, H \in \mathcal{A}$ and every $m \in \mathfrak{g}$, for either sign,*

$$\{F, \{G, H\}\}(m) + \{G, \{H, F\}\}(m) + \{H, \{F, G\}\}(m) = 0.$$

(The outer brackets are defined because $\{G, H\}$ etc. satisfy (A1) by Proposition 6.2; nothing more is needed.)

Proof. Work with the + sign and write $f = \frac{\delta F}{\delta m}(m)$, $g = \frac{\delta G}{\delta m}(m)$, $h = \frac{\delta H}{\delta m}(m)$, $k_f = K_f m$, $k_g = K_g m$, $k_h = K_h m$ (all in $\mathfrak{g}$), and $Q_F = Q_F(m)$ etc.

Step 1 (insert the exact derivative). By definition and (3),

$$\begin{aligned}\{F,\{G,H\}\} &= \langle m, [f, \tfrac{\delta\{G,H\}}{\delta m}]\rangle \\ &= \langle m,[f,[g,h]]\rangle + \langle m,[f,\mathrm{Hess}_G\, k_h]\rangle - \langle m,[f,\mathrm{Hess}_H\, k_g]\rangle.\end{aligned}$$

Step 2 (convert each second-derivative term into a value of a Hessian form). For any $X \in \mathfrak{g}$, antisymmetry and Lemma 5.1 give $\langle m,[f,X]\rangle = -\langle m,[X,f]\rangle = -\langle k_f, X\rangle$. With $X = \mathrm{Hess}_G\, k_h$ and $X = \mathrm{Hess}_H\, k_g$:

$$\begin{aligned}\langle m,[f,\mathrm{Hess}_G\, k_h]\rangle &= -\langle k_f, \mathrm{Hess}_G\, k_h\rangle = -Q_G(k_f,k_h),\\ -\langle m,[f,\mathrm{Hess}_H\, k_g]\rangle &= +\langle k_f, \mathrm{Hess}_H\, k_g\rangle = +Q_H(k_f,k_g).\end{aligned}$$

Therefore

$$\{F,\{G,H\}\} = \langle m,[f,[g,h]]\rangle - Q_G(k_f,k_h) + Q_H(k_f,k_g). \tag{5}$$

Step 3 (the other two terms). Equation (5) holds for every ordered triple of admissible functionals. Applying it to (G,H,F) and to (H,F,G):

$$\{G,\{H,F\}\} = \langle m,[g,[h,f]]\rangle - Q_H(k_g,k_f) + Q_F(k_g,k_h), \tag{6}$$

$$\{H,\{F,G\}\} = \langle m,[h,[f,g]]\rangle - Q_F(k_h,k_g) + Q_G(k_h,k_f). \tag{7}$$

Step 4 (the six second-derivative terms). Adding (5)–(7), the six Hessian terms are, grouped by which functional's Hessian occurs:

Hessian	term and origin		partner and origin		sum
Q_G	$-Q_G(k_f,k_h)$	from (5)	$+Q_G(k_h,k_f)$	from (7)	0
Q_H	$+Q_H(k_f,k_g)$	from (5)	$-Q_H(k_g,k_f)$	from (6)	0
Q_F	$+Q_F(k_g,k_h)$	from (6)	$-Q_F(k_h,k_g)$	from (7)	0

Each row is the same symmetric bilinear form evaluated on the same two vectors in opposite order, so each row sums to zero by Lemma 4.3 (Schwarz). Explicitly, $\big(-Q_G(k_f,k_h) + Q_G(k_h,k_f)\big) + \big(Q_H(k_f,k_g) - Q_H(k_g,k_f)\big) + \big(Q_F(k_g,k_h) - Q_F(k_h,k_g)\big) = 0+0+0$.

Step 5 (what remains).

$$\{F,\{G,H\}\} + \{G,\{H,F\}\} + \{H,\{F,G\}\} = \langle m,[f,[g,h]] + [g,[h,f]] + [h,[f,g]]\rangle = \langle m,0\rangle = 0$$

by Proposition 3.6.

The − sign. By Proposition 6.2, $\delta\{G,H\}_-/\delta m = -\delta\{G,H\}_+/\delta m$, so $\{F,\{G,H\}_-\}_- = -\langle m,[f,-\delta\{G,H\}_+/\delta m]\rangle = \{F,\{G,H\}_+\}_+$; the cyclic sum is the same. □

Remark 7.2 (Where exactly the hard terms go)**.** Each Hessian term is produced by differentiating the *inner* bracket, and each is paired with a term coming from a *different* nested bracket in the cyclic sum: the G-Hessian appears in $\{F,\{G,H\}\}$ (where G is inside, first slot) and in $\{H,\{F,G\}\}$ (where G is inside, second slot), with opposite signs and the two coadjoint vectors k_f, k_h swapped. Two inputs are essential: antisymmetry of the bracket (Step 2, and the sign difference between first and second slot in (3)) and symmetry of second derivatives (Step 4). Dropping the Hessian terms consistently in all three brackets happens to give the right total, but only *because* of Step 4; individually each $\{F,\{G,H\}\}$ changes by an $O(1)$ amount (test (b) below).

8 Vorticity form and correction of the draft's integration by parts

Proposition 8.1 (Vorticity form)**.** *(i) For all $F, G \in \mathcal{A}$ and $m \in \mathfrak{g}$, with $\omega = \operatorname{curl} m$ and $v_F = \frac{\delta F}{\delta m}(m)$, $v_G = \frac{\delta G}{\delta m}(m)$:*

$$\{F, G\}_+(m) = \int_{\mathbb{T}^3} \omega \cdot (v_F \times v_G)\, \mathrm{d}x.$$

(ii) If F factors through vorticity, $F(m) = \Phi(\operatorname{curl} m)$ with $\Phi : \mathfrak{g}_0 \to \mathbb{R}$ having a derivative representative $\frac{\delta \Phi}{\delta \omega}(\omega) \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ ($D\Phi(\omega)\alpha = \langle \alpha, \frac{\delta \Phi}{\delta \omega} \rangle$ for $\alpha \in \mathfrak{g}_0$; it is determined only modulo gradients and constants), then $v_F = \operatorname{curl} \frac{\delta \Phi}{\delta \omega} \in \mathfrak{g}_0$ is well defined, and for two such functionals

$$\{F, G\}_+ = \int_{\mathbb{T}^3} \omega \cdot \Big(\operatorname{curl} \frac{\delta F}{\delta \omega} \times \operatorname{curl} \frac{\delta G}{\delta \omega} \Big)\, \mathrm{d}x,$$

which depends on m only through ω (the mean $\bar{m}$ of Lemma 2.2 drops out).

Proof. (i) is Lemma 5.1 with $v = v_F$, $w = v_G$. (ii) $DF(m)a = D\Phi(\omega)(\operatorname{curl} a) = \langle \operatorname{curl} a, \frac{\delta \Phi}{\delta \omega} \rangle = \langle a, \operatorname{curl} \frac{\delta \Phi}{\delta \omega} \rangle$, using $\int A \cdot \operatorname{curl} B = \int \operatorname{curl} A \cdot B$ (proof of Lemma 5.1). Changing $\frac{\delta \Phi}{\delta \omega}$ by a gradient or a constant does not change its curl. Then apply (i). □

Remark 8.2. Functionals that see the mean momentum, e.g. total momentum $F(m) = \langle m, c \rangle$ with $c \in \mathbb{R}^3$ constant, do not factor through ω; for them (i) still holds with $v_F = c$, e.g. $\{F, G\}_+ = \int \omega \cdot (c \times v_G)$, which is generally nonzero (momentum generates translations; it is not a Casimir). In the Biot–Savart form one writes $m = \bar{m} + \operatorname{curl}(-\Delta)^{-1}\omega$ (Lemma 2.2). The overall sign of the vorticity bracket is convention dependent; ours is fixed by Proposition 5.3.

The draft's step is false. The draft used

$$\int_{\mathbb{T}^3} \omega \cdot \big((v_G \cdot \nabla) v_F\big)\, \mathrm{d}x \;\stackrel{?}{=}\; \int_{\mathbb{T}^3} \omega \cdot (v_F \times v_G)\, \mathrm{d}x. \tag{8}$$

The correct identity (Corollary 3.3 and Lemma 5.1) is

$$\int_{\mathbb{T}^3} m \cdot \big((v_G \cdot \nabla) v_F - (v_F \cdot \nabla) v_G\big)\, \mathrm{d}x = \int_{\mathbb{T}^3} (\operatorname{curl} m) \cdot (v_F \times v_G)\, \mathrm{d}x, \tag{9}$$

with three differences from (8): the left side pairs with the momentum m, not with $\omega = \operatorname{curl} m$; *both* transport terms are needed; and the curl is moved from $v_F \times v_G$ onto m by integration by parts. *Counterexample to* (8) (exact, sympy): on $\mathbb{T}^3$ take the divergence-free trigonometric fields

$$\begin{aligned}
m = \big(& \sin y + \sin(x+y) - \sin(y+z) - \cos y - \cos(x+y), \\
& - \sin(x+y) + \sin(y+z) - \cos x + \cos(x+y) + \cos(y+z), \\
& - \sin x - \sin y + 2\sin(x+y) - \sin(y+z) + \cos y + 2\cos(x+y) - \cos(y+z)\big), \\
v_F = \big(& - \sin y + \sin(x+z) + \cos(x+y+z), \quad - \sin(x+y+z) + 2\cos(x+z), \\
& - \sin(x+z) + \sin(x+y+z) + \cos y - \cos(x+y+z)\big), \\
v_G = \big(& - \sin(x-y) + \sin(y+z) + \sin(x+y+z) + \cos(x-y) + 2\cos(y+z) - \cos(x+y+z), \\
& \sin x - \sin(x-y) + \cos x + \cos(x-y) + \cos(x+y+z), \\
& 2\sin x - \sin(x+y+z) + 2\cos(x-y)\big),
\end{aligned}$$

$(x, y, z) = (x_1, x_2, x_3)$, $\omega = \operatorname{curl} m$. Then

$$\int \omega \cdot ((v_G \cdot \nabla) v_F)\, \mathrm{d}x = -4\pi^3, \qquad \int \omega \cdot (v_F \times v_G)\, \mathrm{d}x = +4\pi^3,$$

so (8) fails, while both sides of (9) equal $4\pi^3$ and the single transport term $\int m \cdot ((v_G \cdot \nabla) v_F)\, \mathrm{d}x = 8\pi^3$ differs from both. The draft also identified $\mathfrak{g}^*$ with "vorticity fields" through the pairing $\int \omega \cdot v$; under that pairing the dual variable plays the role of the momentum m, not of $\operatorname{curl} m$, and mixing the two roles produces exactly this kind of error.

9 Computational checks

All scripts are in `scripts/`, outputs in `outputs/`. These checks are sanity checks, not proofs; the proofs are Sections 3–8.

Symbolic (`sympy_identities.py`, sympy 1.14). With *generic* (undetermined) smooth component functions: Lemma 3.2; $[v, w] = \operatorname{curl}(v \times w)$ and $\operatorname{div}[v, w] = 0$ for $v = \operatorname{curl} A$, $w = \operatorname{curl} B$; the vector-field Jacobi identity for arbitrary fields; the pointwise identity $m \cdot \operatorname{curl} c - \operatorname{curl} m \cdot c = \operatorname{div}(c \times m)$. With exact trigonometric fields: (9), the counterexample to (8), Lemma 5.1, and the sign check of Proposition 5.3. Finite-dimensional sanity check: Lie–Poisson Jacobi on $\mathfrak{so}(3)^*$ and $\mathfrak{sl}(2, \mathbb{R})^*$ for *arbitrary* smooth F, G, H (sympy undetermined functions), where a single $\{F, \{G, H\}\}$ contains 66 terms with second derivatives that cancel only in the cyclic sum.

```
[PASS] curl(a x b) = a div b - b div a + (b.grad)a - (a.grad)b  (generic smooth a,b)
[PASS] div(curl A) = 0
[PASS] [v,w] = curl(v x w) for v=curl A, w=curl B (div-free)
[PASS] closure: div [v,w] = 0 for div-free v,w
   non-div-free example v=(sin x,0,0), w=(0,cos x,0): [v,w]-curl(v x w) = [0, cos(x)**2, 0]
[PASS] vector-field Jacobi [u,[v,w]]+[v,[w,u]]+[w,[u,v]] = 0 (generic fields)
[PASS] m.curl(c) - curl(m).c = div(c x m), c = v x w (generic)
   <m,[vF,vG]> = 4*pi**3,   int omega.(vF x vG) = 4*pi**3
[PASS] correct identity <m,[vF,vG]> = int curl(m).(vF x vG) on concrete trig fields
   draft step: int omega.((vG.grad)vF) = -4*pi**3  vs  int omega.(vF x vG) = 4*pi**3
   (also: int m.((vG.grad)vF) alone = 8*pi**3; the second transport term is needed)
[PASS] counterexample: the draft's integration-by-parts step is false on these fields
[PASS] <m,[vF,vG]> = int (vG x omega).vF  (K_w m = P(w x omega))
   {F,H}_+ = -2*pi**3,  -<(u.grad)u,a> = -2*pi**3
[PASS] + sign reproduces Euler: {<.,a>, 1/2|.|^2}_+ (u) = -<(u.grad)u, a>
[PASS] finite-dim Lie-Poisson Jacobi on so(3)* for arbitrary smooth F,G,H
   so(3): {F,{G,H}} alone has 66 terms containing second derivatives (nonzero individually)
[PASS] finite-dim Lie-Poisson Jacobi on sl(2,R)* for arbitrary smooth F,G,H
   sl(2,R): {F,{G,H}} alone has 66 terms containing second derivatives (nonzero individually)
ALL PASS
```

Exact-quadrature floating point on $\mathbb{T}^3$ (`torus_fft_check.py`). A Galerkin truncation of $\mathfrak{g}$ is *not* closed under the bracket, so we do not truncate the bracket; instead all fields are exact trigonometric polynomials (degree ≤ 2 for the random inputs), products are formed pointwise, and a 48^3 grid is used, on which spectral differentiation, the Leray projection and quadrature are exact for every trigonometric polynomial that occurs (max degree $\leq 16 < 24$ where derivatives or P are applied; integrands of degree $\leq 30 < 48$). Remaining error is roundoff, plus finite-difference error in test (a). Functionals: F cylindrical with $\Phi(y) = y_1 y_2^2 + \sin y_3 + y_1 y_3$; $G(m) = \int m_1 m_2 m_3\, \mathrm{d}x$ (local cubic); H cylindrical with $\Phi(y) = e^{0.1 y_1} y_2 + y_3^3$; $E = \frac{1}{2}\langle m, m \rangle$; m random with nonzero mean. (0) checks the ingredients; (a) compares a 6th-order finite-difference derivative of $\{G, H\}$ with $\langle a, (3) \rangle$

(agreement $\sim 10^{-13}$; $[g,h]$ alone is off by 29–93%); (b) evaluates the three nested brackets via (3), the six Hessian terms of Step 4 and their pair sums ($\sim 10^{-16}$), and the Jacobiator term.

```
N=48; max |div m| = 5.36e-15; mean of m = [-0.0199  0.0005  0.0247]
(0a) |<m,[v,w]> - <Kop(w,m),v>|          = 1.1102230246251565e-16
(0b) |<m,[v,w]> - int curl m.(v x w)|     = 1.1102230246251565e-16
(0c) max|div [v,w]|                        = 4.579669976578771e-14
(0d) max|vector-field Jacobi|              = 1.73749903353837e-14  (scale 0.7150610186093074 )
(0e) Hessian symmetry F: |<a,Hb>-<b,Ha>| = 3.552713678800501e-15
(0e) Hessian symmetry G: |<a,Hb>-<b,Ha>| = 0.0
(0e) Hessian symmetry H: |<a,Hb>-<b,Ha>| = 0.0
(a) {G,H}: FD derivative=6.7238193316e-01  <a,formula>=6.7238193316e-01  rel.err=6.0e-13   | naive <a,[dA,dB]>
     only = 9.343466e-01 (rel.err 3.9e-01)
(a) {F,G}: FD derivative=8.0323432620e-01  <a,formula>=8.0323432620e-01  rel.err=2.0e-13   | naive <a,[dA,dB]>
     only = 5.715599e-01 (rel.err 2.9e-01)
(a) {H,E}: FD derivative=3.8271323414e+00  <a,formula>=3.8271323414e+00  rel.err=6.0e-14   | naive <a,[dA,dB]>
     only = 7.369127e+00 (rel.err 9.3e-01)
(b) F,G,H: terms = ['2.464242e-01', '-3.134420e+00', '2.887996e+00']  cyclic sum = -4.441e-16  (rel 1.4e-16)
    per-term effect of the second-derivative terms (full - naive): ['9.835317e-01', '-2.973897e+00', '1.990365
     e+00']
    naive terms (second-derivative terms dropped): ['-7.371075e-01', '-1.605232e-01', '8.976307e-01'], their
     cyclic sum = 5.6e-16
    -> each individual {A,{B,C}} changes by an O(1) amount; only the cyclic sum of the six omitted terms
     vanishes
    six second-derivative terms: {'-Q_B(kA,kC)': '1.183036e-01', '+Q_B(kC,kA)': '-1.183036e-01', '+Q_C(kA,kB)
     ': '8.652281e-01', '-Q_C(kB,kA)': '-8.652281e-01', '+Q_A(kB,kC)': '-2.108669e+00', '-Q_A(kC,kB)':
     '2.108669e+00'}
    pair sums: B:-3.9e-16  C:-1.1e-16  A:4.4e-16
    <m, [a,[b,c]]+cyc> = 2.3e-16
(b) F,G,E: terms = ['-1.407024e-03', '1.677543e-02', '-1.536841e-02']  cyclic sum = 1.579e-16  (rel 9.4e-15)
    per-term effect of the second-derivative terms (full - naive): ['2.494547e-01', '-1.526149e-01',
     '-9.683979e-02']
    naive terms (second-derivative terms dropped): ['-2.508617e-01', '1.693903e-01', '8.147138e-02'], their
     cyclic sum = 9.7e-17
    -> each individual {A,{B,C}} changes by an O(1) amount; only the cyclic sum of the six omitted terms
     vanishes
    six second-derivative terms: {'-Q_B(kA,kC)': '1.328377e-01', '+Q_B(kC,kA)': '-1.328377e-01', '+Q_C(kA,kB)
     ': '1.166170e-01', '-Q_C(kB,kA)': '-1.166170e-01', '+Q_A(kB,kC)': '-3.599792e-02', '-Q_A(kC,kB)':
     '3.599792e-02'}
    pair sums: B:-2.8e-17  C:0.0e+00  A:-6.9e-18
    <m, [a,[b,c]]+cyc> = 7.7e-17
(b) G,H,E: terms = ['1.788916e-02', '3.135361e-01', '-3.314252e-01']  cyclic sum = -1.110e-16  (rel 3.3e-16)
    per-term effect of the second-derivative terms (full - naive): ['1.276241e-01', '2.543003e-01', '-3.819244
     e-01']
    naive terms (second-derivative terms dropped): ['-1.097350e-01', '5.923583e-02', '5.049913e-02'], their
     cyclic sum = -1.1e-16
    -> each individual {A,{B,C}} changes by an O(1) amount; only the cyclic sum of the six omitted terms
     vanishes
    six second-derivative terms: {'-Q_B(kA,kC)': '3.343983e-01', '+Q_B(kC,kA)': '-3.343983e-01', '+Q_C(kA,kB)
     ': '-2.067742e-01', '-Q_C(kB,kA)': '2.067742e-01', '+Q_A(kB,kC)': '4.752609e-02', '-Q_A(kC,kB)':
     '-4.752609e-02'}
    pair sums: B:1.1e-16  C:0.0e+00  A:-7.6e-17
    <m, [a,[b,c]]+cyc> = -8.6e-17
```

10 Lean 4 formalization status

What is formalized. `FTVerified/Companion/LiePoissonJacobiCore.lean` in `https://github.com/BenFrohman/frohmanian-tether-verified` (code commit `cf04afd`, pull request #2) proves, for an arbitrary Lie algebra L over a commutative ring R, any linear $m : L \to R$, any

$f, g, h \in L$, any k_f, k_g, k_h in a type V and any symmetric $Q_F, Q_G, Q_H : V \to V \to R$, that

$$\big(m[f,[g,h]] - Q_G(k_f,k_h) + Q_H(k_f,k_g)\big) + \big(m[g,[h,f]] - Q_H(k_g,k_f) + Q_F(k_g,k_h)\big) + \big(m[h,[f,g]] - Q_F(k_h,k_g) + Q_G(k_h,k_f)\big) = 0,$$

i.e. exactly Steps 4–5 of Theorem 7.1, using mathlib's `lie_jacobi`. It is checked with Lean `v4.30.0-rc1` and mathlib commit `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7` by the repository's GitHub Actions workflow (run `https://github.com/BenFrohman/frohmanian-tether-verified/actions/runs/42923330722`, passed), which builds the library and fails on any non-standard axiom. Its output for this file:

```
'LiePoissonCore.six_terms_cancel' depends on axioms: [propext, Quot.sound]
'LiePoissonCore.lie_poisson_jacobi_core' depends on axioms: [propext, Quot.sound]
```

There is no `sorry`, `admit`, `native_decide` or custom axiom. The hypotheses of `lie_poisson_jacobi_core` encode exactly the inputs of Steps 4–5: m linear (the pairing with the state), Q_F, Q_G, Q_H symmetric (Lemma 4.3, symmetry of second derivatives), and the Lie algebra axioms of L. The symmetry hypotheses are satisfiable (for example $Q = 0$), so the statement is not vacuous; and it is not trivially true without them, since the six Q terms cancel only by symmetry.

What is not formalized. Everything analytic: the torus, $\mathfrak{g}$, the Leray projection, the admissible class, Lemma 4.3, Lemma 6.1 and Proposition 6.2. A Lean formalization would need (a) in finite dimensions: $F : \mathrm{Dual}\,\mathbb{R}\,\mathfrak{g} \to \mathbb{R}$ smooth, gradients via finite-dimensional duality, the derivative of $m \mapsto m([\nabla F(m), \nabla G(m)])$ (product/chain rule for `fderiv` of bilinear maps), and symmetry of the second derivative (available in mathlib as `IsSymmSndFDerivAt` results for C^2 maps) — feasible but not done here; (b) for $\mathfrak{sdiff}(\mathbb{T}^3)$: smooth periodic vector fields on (AddCircle)^3, integration by parts, the Fourier/Leray projection on C^∞, and the Gateaux/Fréchet setting of Definition 4.2 — a substantial project, not attempted.

11 Scope, honesty statement and caveats

- **Classical, not new.** The Lie–Poisson structure of ideal hydrodynamics and its Jacobi identity are due to Arnold [1] and Marsden–Weinstein [2]; see also Arnold–Khesin [3] and Marsden–Ratiu [4]. This note re-derives the identity with every cancellation displayed; it claims no new result.
- **Lie–Poisson part only.** The theorem covers $\{F, G\} = \pm\langle m, [\delta F, \delta G]\rangle$ and nothing else. Any added symmetric (metriplectic/"tether") bracket T, or any dissipative bracket, is not covered; in metriplectic systems (Morrison [5]) the symmetric part is not a Lie bracket, so no Jacobi identity is asserted for it, and the sum of the Lie–Poisson and symmetric parts is not a Poisson bracket. Likewise, finite Fourier (Galerkin or frozen-coefficient) models are different objects and are not covered.
- **No PDE or Clay claim.** Nothing here concerns existence, regularity, or blow-up for the Euler or Navier–Stokes equations.
- **Function class.** The proof is for the admissible class $\mathcal{A}$ of Definition 4.2 on the regular dual, in a C^∞ Gateaux setting with assumptions (A1)–(A3). It does not cover functionals whose derivative is a distribution, nor rough (H^s, L^2) phase spaces.

- **Torus, not $\mathbb{R}^3$.** All statements are on $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$; the constant (harmonic) fields are included in $\mathfrak{g}$ and handled explicitly; vorticity formulas lose the mean momentum.
- **Draft.** Only an excerpt of the earlier draft was available; the corrections in Sections 8 and 3 refer to the steps quoted there.
- **Numerics.** Section 9 gives evidence for the formulas, not proofs.
- **Lean.** Only the final algebraic cancellation is machine-checked (Section 10).

Reproduction. `python3 scripts/sympy_identities.py` (about 3–4 minutes), `python3 scripts/torus_fft_check.py` (about 10 s; NumPy, fixed seed 20261109), for the Lean part, in the repository root, `lakeexecacheget&&lakebuild&&python3scripts/check_jacobi_axioms.py`, and `tectonic lie_poisson_jacobi.tex`.

References

[1] V. I. Arnold, Sur la géométrie différentielle des groupes de Lie de dimension infinie et ses applications à l'hydrodynamique des fluides parfaits, *Ann. Inst. Fourier* **16** (1966), 319–361.

[2] J. Marsden and A. Weinstein, Coadjoint orbits, vortices, and Clebsch variables for incompressible fluids, *Physica D* **7** (1983), 305–323.

[3] V. I. Arnold and B. A. Khesin, *Topological Methods in Hydrodynamics*, Springer, 1998 (2nd ed. 2021).

[4] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Springer, 1999.

[5] P. J. Morrison, A paradigm for joined Hamiltonian and dissipative systems, *Physica D* **18** (1986), 410–419.